# A Vanishing-Cycle Dehn Twist for the Hironaka–Koch $\mathrm{Per}_4(0)$ Deformation Space: Preservation, Base-Component Triviality, and the Open Infinite-Orbit Problem

**Abstract**

We give a partial answer to the problem of exhibiting an explicit simple closed curve whose Dehn-twist power preserves the Hironaka–Koch deformation space attached to a quadratic rational map with a superattracting 4-cycle and one free critical value, and whose action on the connected components has an infinite orbit. For the concrete Hironaka–Koch map

$$f(z) = \frac{(4z-3)(z+2)}{4z^2},$$

with superattracting cycle $0 \mapsto \infty \mapsto 1 \mapsto \frac{3}{4} \mapsto 0$ and free critical value $v = 121/96$, let $\gamma_{0,v}$ be any essential simple closed curve separating $\{0, v\}$ from $\{\infty, 1, \frac{3}{4}\}$. We prove that the square $T^2_{\gamma_{0,v}}$ of the Dehn twist about this curve preserves the deformation space $\mathcal{D}_f$. We also prove, using the base-point-component form of the Hironaka–Koch monodromy subgroup $E_f$, that $T^2_{\gamma_{0,v}} \in E_f$, so it fixes the base connected component of $\mathcal{D}_f$. Thus the explicit curve $\gamma_{0,v}$ and the power 2 solve the preservation half of the problem, but they do not supply the requested infinite orbit. The known Hironaka–Koch infinite-component mechanism is produced by a braid element that is not identified here with a Dehn twist about a single simple closed curve.

## 1 Introduction

Let $f : \mathbb{S}^2 \to \mathbb{S}^2$ be a branched cover, let $V(f)$ denote its set of critical values, and let $A, B \subset \mathbb{S}^2$ be finite sets. Whenever $f(A) \cup V(f) \subset B$, pulling back complex structures on $(\mathbb{S}^2, B)$ yields a holomorphic map

$$f^* : \mathrm{Teich}(B) \longrightarrow \mathrm{Teich}(A),$$

and whenever $A \subset B$, forgetting the points $B \setminus A$ yields

$$i_* : \mathrm{Teich}(B) \longrightarrow \mathrm{Teich}(A).$$

When $f(A) \cup V(f) \cup A \subset B$, the deformation space is the equalizer

$$\mathcal{D}_f = \{\tau \in \mathrm{Teich}(B) : f^*(\tau) = i_*(\tau)\}.$$

The original problem asks for an explicit simple closed curve $\gamma \subset (\mathbb{S}^2, B)$ such that some power of the Dehn twist $T_\gamma$ preserves $\mathcal{D}_f$ and acts on the set of connected components $\pi_0(\mathcal{D}_f)$ with an infinite orbit. We work with the specific quadratic rational map used by Hironaka and Koch in [1, §4]:

$$f(z) = \frac{(4z-3)(z+2)}{4z^2}. \tag{1.1}$$

Its critical points are 0 and 12/5. The point 0 lies in the superattracting 4-cycle

$$A = \left\{0, \infty, 1, \tfrac{3}{4}\right\}, \qquad 0 \longmapsto \infty \longmapsto 1 \longmapsto \tfrac{3}{4} \longmapsto 0,$$

while the second critical value is the free critical value

$$v = f\left(\tfrac{12}{5}\right) = \frac{121}{96}.$$

We set

$$B = A \cup \{v\}.$$

Let $\gamma_{0,v}$ be any essential simple closed curve in $\mathbb{S}^2 \setminus B$ bounding a disk whose intersection with $B$ is exactly $\{0, v\}$: it separates $\{0, v\}$ from $\{\infty, 1, \frac{3}{4}\}$. Geometrically, $\gamma_{0,v}$ is the vanishing cycle of the collision of the free critical value $v$ with the periodic critical point 0.

The present paper proves the following two results, and records honestly what is not proved.

**Theorem A (preservation).** For each $a \in \{0, 1, \frac{3}{4}\}$, if $\gamma_{a,v}$ is any essential simple closed curve separating $\{a, v\}$ from the other three points of $B$, then $T^2_{\gamma_{a,v}} \in S_f$, and consequently

$$T^2_{\gamma_{a,v}} \cdot \mathcal{D}_f = \mathcal{D}_f.$$

In particular, $\gamma = \gamma_{0,v}$ satisfies the preservation requirement of the problem with power 2.

**Theorem B (base-component triviality).** With the Hironaka–Koch base point and marking conventions, the vanishing-cycle twist satisfies

$$T^2_{\gamma_{0,v}} \in E_f,$$

where $E_f$ is the image of the monodromy of the Hironaka–Koch parameter space $V = \mathrm{Per}_4(0)^*$. Consequently, if $D_0$ is the connected component of $\mathcal{D}_f$ containing the base point $\tau_*$, then $T^2_{\gamma_{0,v}}(D_0) = D_0$.

Thus the same explicit curve $\gamma = \gamma_{0,v}$ that supplies the preservation half of the problem does not supply an infinite orbit: its square lies in $E_f$ and fixes the base component. The infinite-orbit requirement for a Dehn twist about a single simple closed curve remains open. The known Hironaka–Koch infinite-component phenomenon is produced by the braid element $g = s_2 c r_1 s_1 c r_2$ of [1, Theorem 1.3 and Remark 4.7], which we cite but do not identify with a Dehn twist. The paper is organised as follows. Section 2 recalls the Hironaka–Koch framework, states their stabiliser criterion, and derives the base-component lemma we use. Section 3 proves the normal-form algebra for quadratic rational maps with a superattracting 4-cycle and computes the multiplicity-two collision between $v$ and the periodic critical point. Section 4 verifies the concrete data of (1.1), establishes the structure of the preimage of the vanishing cycle, and proves Theorem A. Section 5 proves Theorem B. Section 6 summarises the consequences and records the precise open points.

# 2 Preliminaries and Hironaka–Koch input

Throughout, $\mathbb{S}^2$ is the Riemann sphere. For a finite set $P \subset \mathbb{S}^2$ with $|P| \geq 3$, let $\mathrm{Teich}(P)$ denote the Teichmüller space of the marked sphere $(\mathbb{S}^2, P)$, and let

$$\mathrm{Mod}_P = \pi_0 \,\mathrm{Homeo}^+(\mathbb{S}^2, P)$$

denote the pure mapping class group: orientation-preserving homeomorphisms fixing $P$ pointwise, modulo isotopy through such homeomorphisms. For an essential simple closed curve $\gamma \subset \mathbb{S}^2 \setminus P$, we write $T_\gamma$ for the right Dehn twist about $\gamma$.

If $Q \subset P$ has cardinality 2, we let $\gamma_Q$ denote any essential simple closed curve in $\mathbb{S}^2 \setminus P$ bounding a disk $D$ with $D \cap P = Q$. Such a curve separates $\mathbb{S}^2$ into a disk containing $Q$ and a disk containing $P \setminus Q$. For the concrete map of Section 4, we use the notation

$$A = \{0, \infty, 1, \tfrac{3}{4}\}, \qquad B = A \cup \{v\}, \qquad v = \tfrac{121}{96},$$

and for $a \in A$ we write $\gamma_{a,v}$ for $\gamma_{\{a,v\}}$.

## 2.1 The Hironaka–Koch deformation setup

We work inside the framework of Hironaka–Koch [1, §2.5–§2.6]. A mapping class $g \in \mathrm{Mod}_B$ is called *$f$-liftable* if there exists a class $\Phi_f(g) \in \mathrm{Mod}_A$ such that

$$f^*(g \cdot \tau) = \Phi_f(g) \cdot f^*(\tau) \qquad \text{for all } \tau \in \mathrm{Teich}(B).$$

Equivalently, $g$ admits a lift to a homeomorphism of $(\mathbb{S}^2, f^{-1}(B))$ fixing $A$ pointwise after forgetting $f^{-1}(B) \setminus A$. The liftable classes form a subgroup $L_f \subset \mathrm{Mod}_B$, and the lifting homomorphism is

$$\Phi_f : L_f \longrightarrow \mathrm{Mod}_A\,.$$

Because $A \subset B$, forgetting $B \setminus A$ gives a homomorphism

$$\Phi_A^B : \mathrm{Mod}_B \longrightarrow \mathrm{Mod}_A\,.$$

The *special liftable* subgroup is the equalizer

$$S_f = \{g \in L_f : \Phi_f(g) = \Phi_A^B(g)\}.$$

We use the following standard Hironaka–Koch criterion. We cite it rather than reprove it, and we check its hypotheses for our concrete data in Section 4.

**Proposition 2.1** (Hironaka–Koch [1, Proposition 2.1])**.** *Let $f : \mathbb{S}^2 \to \mathbb{S}^2$ be a branched cover and let $A, B \subset \mathbb{S}^2$ be finite sets with $A \subset B$, $|A| \geq 3$, $|B| \geq 3$, and*

$$f(A) \cup V(f) \cup A \subset B.$$

*With $L_f, \Phi_f, \Phi_A^B, S_f$ as above, if $g \in L_f$ then the following are equivalent:*

*(i) $g \cdot \mathcal{D}_f \cap \mathcal{D}_f \neq \varnothing$;*

*(ii)* $g \cdot \mathcal{D}_f = \mathcal{D}_f$;

*(iii)* $g \in S_f$.

*Proof.* This is [1, Proposition 2.1]. The hypotheses used there are exactly that both Teichmüller spaces are non-exceptional ($|A|, |B| \geq 3$) so that $f^*$ and $i_*$ are well defined, and that $f(A) \cup V(f) \cup A \subset B$ so that the deformation space $\mathcal{D}_f = \{\tau \in \mathrm{Teich}(B) : f^*(\tau) = i_*(\tau)\}$ is meaningful.

For the map $f$ of (1.1), Lemma 4.1 below computes $f(A) = A$ and $V(f) = \{\infty, v\}$, so

$$f(A) \cup V(f) \cup A = A \cup \{v\} = B,$$

and $|A| = 4$, $|B| = 5$. Hence the hypotheses of the cited proposition are satisfied. □

## 2.2 The Hironaka–Koch parameter space and the subgroup $E_f$

Let

$$V = \mathrm{Per}_4(0)^*$$

be the Hironaka–Koch parameter space of quadratic rational maps with a marked superattracting 4-cycle and one free critical value, in the normalization of [1, §4]. Let $*_V \in V$ be the base point corresponding to the concrete map (1.1), and let $\tau_* \in \mathrm{Teich}(B)$ be the associated marked point. Hironaka–Koch construct a monodromy homomorphism

$$\rho : \pi_1(V, *_V) \longrightarrow S_f$$

and define

$$E_f = \rho(\pi_1(V, *_V)).$$

In [1] this subgroup is written $i_*(\pi_1(V, *_V))$; to avoid a clash of notation with the forgetful map $i_* : \mathrm{Teich}(B) \to \mathrm{Teich}(A)$, we write $\rho$ for the monodromy homomorphism.

We shall not use the full coset description of the connected components of $\mathcal{D}_f$ from [1, Proposition 3.4]. All we need is the following base-point-component consequence.

**Lemma 2.2.** *Let $D_0$ be the connected component of $\mathcal{D}_f$ containing the base point $\tau_*$. If $g \in E_f$, then $g(D_0) = D_0$.*

*Proof.* Choose a loop $\ell$ in $V$ based at $*_V$ with $\rho([\ell]) = g$. By the Hironaka–Koch construction of the monodromy action, the basepoint marking transported along the loop gives a path of marked structures on which the deformation equation holds at every parameter; thereore the lifted path lies in $\mathcal{D}_f$. Starting at $\tau_*$ the endpoint of this path is $g \cdot \tau_*$ (in the action convention of [1]), so $g \cdot \tau_* \in D_0$.

Since $E_f \subset S_f$, Proposition 2.1 gives $g \cdot \mathcal{D}_f = \mathcal{D}_f$. Hence $g(D_0)$ is a connected component of $\mathcal{D}_f$. This component contains $g \cdot \tau_*$, and $D_0$ also contains $g \cdot \tau_*$. Since distinct components are disjoint, $g(D_0) = D_0$. □

*Remark* 2.3. The conclusion of Lemma 2.2 is deliberately limited to the base component. If $E_f$ were known to be normal in $S_f$, one could draw further conclusions about the action on all components; the arguments of this paper do not establish normality, and we do not rely on it.

# 3 Normal form and the double collision

The next theorem records the one-parameter normal form for quadratic rational maps with a superattracting 4-cycle, computes the second critical point and its value, and identifies the collision between the free critical value and the periodic critical point together with its local multiplicity.

**Theorem 3.1.** *For $\alpha$ away from the exceptional loci on which the formulas below degenerate, define*

$$f_\alpha(z) = \frac{\alpha(z-\alpha)^2}{(z-1)(z-\alpha^3)}.$$

*Then:*

*(i) $\alpha$ is a critical point, and*

$$\alpha \longmapsto 0 \longmapsto 1 \longmapsto \infty \longmapsto \alpha$$

*is a superattracting 4-cycle.*

*(ii) The second critical point is*

$$c_2(\alpha) = -\frac{\alpha(\alpha^2-\alpha-1)}{\alpha^2+\alpha-1}.$$

*(iii) Its critical value is*

$$v(\alpha) = f_\alpha(c_2(\alpha)) = \frac{4\alpha^2(\alpha+1)}{(\alpha^2+\alpha+1)^2}.$$

*(iv) The difference of the free critical value from the periodic critical point factors as*

$$v(\alpha) - \alpha = -\frac{\alpha(\alpha^2+\alpha-1)^2}{(\alpha^2+\alpha+1)^2}.$$

*Hence $v(\alpha) = \alpha$ exactly when $\alpha^2+\alpha-1=0$, and at such parameters the collision has multiplicity 2; additionally $c_2(\alpha) \to \infty = f_\alpha^3(\alpha)$.*

*(v) If $\gamma_\alpha$ is any simple closed curve separating $\{\alpha, v(\alpha)\}$ from $\{0,1,\infty\}$, then the local monodromy of the family $f_\alpha$ around the collision parameter is $T_{\gamma_\alpha}^2$, up to the usual orientation convention for Dehn twists.*

*Proof.* Set

$$N(z) = \alpha(z-\alpha)^2, \qquad D(z) = (z-1)(z-\alpha^3),$$

so that $f_\alpha = N/D$.

*(i).* The numerator has a double zero at $z=\alpha$, so $f$ attains the value 0 there to local degree 2, and $\alpha$ is a critical point. Direct evaluation gives

$$f_\alpha(0) = \frac{\alpha(-\alpha)^2}{(-1)(-\alpha^3)} = 1, \qquad f_\alpha(\infty) = \alpha,$$

while $D(1) = 0$ and $N(1) \neq 0$ away from exceptional $\alpha$, so $f_\alpha(1) = \infty$. Hence

$$\alpha \longmapsto 0 \longmapsto 1 \longmapsto \infty \longmapsto \alpha$$

is a 4-cycle, and it is superattracting because $\alpha$ is critical.

*(ii).* Differentiating,

$$f'_\alpha(z) = \frac{N'(z)D(z) - N(z)D'(z)}{D(z)^2}, \qquad N'(z) = 2\alpha(z-\alpha),$$

so

$$f'_\alpha(z) = \frac{\alpha(z-\alpha)\big(2D(z) - (z-\alpha)D'(z)\big)}{D(z)^2}.$$

One critical point is $z = \alpha$. Writing $D(z) = z^2 - (1+\alpha^3)z + \alpha^3$ and $D'(z) = 2z - (1+\alpha^3)$, we compute

$$\begin{aligned} 2D(z) - (z-\alpha)D'(z) &= 2\big(z^2 - (1+\alpha^3)z + \alpha^3\big) - (z-\alpha)\big(2z - (1+\alpha^3)\big) \\ &= -(\alpha-1)\Big((\alpha^2+\alpha-1)z + \alpha(\alpha^2-\alpha-1)\Big). \end{aligned}$$

The remaining zero of this factor yields the second critical point

$$c_2(\alpha) = -\frac{\alpha(\alpha^2-\alpha-1)}{\alpha^2+\alpha-1}.$$

*(iii).* Set $\beta = c_2(\alpha)$. At the critical point, the equation $2D(\beta) = (\beta-\alpha)D'(\beta)$ holds, hence

$$v(\alpha) = f_\alpha(\beta) = \frac{\alpha(\beta-\alpha)^2}{D(\beta)} = \frac{2\alpha(\beta-\alpha)}{D'(\beta)}.$$

Now

$$\beta - \alpha = -\frac{\alpha(\alpha^2-\alpha-1)}{\alpha^2+\alpha-1} - \alpha = -\frac{2\alpha(\alpha^2-1)}{\alpha^2+\alpha-1},$$

and

$$D'(\beta) = 2\beta - (1+\alpha^3) = -\frac{(\alpha-1)(\alpha^2+\alpha+1)^2}{\alpha^2+\alpha-1}.$$

Substituting,

$$v(\alpha) = \frac{2\alpha \cdot \left(-\dfrac{2\alpha(\alpha^2-1)}{\alpha^2+\alpha-1}\right)}{-\dfrac{(\alpha-1)(\alpha^2+\alpha+1)^2}{\alpha^2+\alpha-1}} = \frac{4\alpha^2(\alpha+1)}{(\alpha^2+\alpha+1)^2}.$$

*(iv).* Combining (iii) with the common denominator,

$$\begin{aligned} v(\alpha) - \alpha &= \frac{4\alpha^2(\alpha+1) - \alpha(\alpha^2+\alpha+1)^2}{(\alpha^2+\alpha+1)^2} \\ &= \frac{-\alpha\big(\alpha^4 + 2\alpha^3 - \alpha^2 - 2\alpha + 1\big)}{(\alpha^2+\alpha+1)^2} \\ &= -\frac{\alpha(\alpha^2+\alpha-1)^2}{(\alpha^2+\alpha+1)^2}. \end{aligned}$$

Thus $v(\alpha) = \alpha$ if and only if $\alpha^2 + \alpha - 1 = 0$. At such $\alpha$, $\alpha^2 + \alpha + 1 = 2$, so the prefactor is nonzero, and since $\alpha^2 + \alpha - 1$ has simple zeros, the collision has multiplicity 2.

At a root of $\alpha^2 + \alpha - 1$, the denominator $\alpha^2 + \alpha - 1$ of $c_2(\alpha)$ vanishes. Using $\alpha^2 = 1 - \alpha$,

$$\alpha^2 - \alpha - 1 = (1 - \alpha) - \alpha - 1 = -2\alpha,$$

so the numerator of $c_2$ tends to $2\alpha^2 \neq 0$ because $\alpha \neq 0$ on the collision locus; hence $c_2(\alpha) \to \infty$. Since $f_\alpha^3(\alpha) = \infty$ by (i), the limiting second critical point coincides with the cycle point $f_\alpha^3(\alpha)$.

*(v).* Let $\alpha_0$ be a root of $\alpha^2 + \alpha - 1$ and write $\alpha - \alpha_0 = t$. By (iv),

$$v(\alpha) - \alpha = u(t)\, t^2, \qquad u(0) \neq 0.$$

Along the loop $t = \varepsilon e^{i\theta}$, $0 \leq \theta \leq 2\pi$, the relative coordinate $v(\alpha) - \alpha$ winds twice around 0. Choose a closed disk $U$ containing $\alpha$ and $v(\alpha)$ and disjoint from $\{0, 1, \infty\}$. Inside $U$, the two marked points move as the pure braid with relative winding number two. Under the standard braid-to-mapping-class dictionary, in the twice-punctured disk a pure braid with relative winding number one represents the Dehn twist about the simple closed curve that surrounds both punctures, and relative winding number two represents its square. See Farb–Margalit [2, Chapter 9, Theorem 9.1 and §9.1] for the identification of the pure braid group on a punctured disk with a subgroup of the mapping class group. Extending by the identity outside $U$ gives the local monodromy $T_{\gamma_\alpha}^2$, up to orientation. □

*Remark* 3.2. The normalization of Hironaka–Koch [1, §4] places the periodic critical point at the value 0 by a Möbius change of coordinates. In that coordinate, the curve $\gamma_\alpha$ of Theorem 3.1 becomes the curve $\gamma_{0,v}$ that separates $\{0, v\}$ from the other three points of $B$.

# 4 The concrete map and liftable squared twists

We now specialize to the map (1.1),

$$f(z) = \frac{(4z - 3)(z + 2)}{4z^2} = \frac{4z^2 + 5z - 6}{4z^2}.$$

Its superattracting 4-cycle is

$$A = \left\{0, \infty, 1, \tfrac{3}{4}\right\},$$

and its free critical value is

$$v = \tfrac{121}{96}, \qquad B = A \cup \{v\}.$$

**Lemma 4.1.** *For the map $f$ of* (1.1)*, the following hold.*

*(i) The cycle is*

$$0 \longmapsto \infty \longmapsto 1 \longmapsto \tfrac{3}{4} \longmapsto 0.$$

*(ii) The critical points are* 0 *and* 12/5*, and the branch values are* $\infty$ *and* $v = 121/96$*.*

*(iii) The nontrivial deck transformation of the degree-two map is*

$$\tau(z) = \frac{6z}{5z-6},$$

*and* $\tau^2 = \mathrm{id}$ *and* $f \circ \tau = f$.

*(iv) The fibers over* $B$ *are*

$$\begin{aligned} f^{-1}(0) &= \left\{\tfrac{3}{4}, -2\right\}, \\ f^{-1}(\infty) &= \{0\} \quad \textit{with branching}, \\ f^{-1}(1) &= \left\{\tfrac{6}{5}, \infty\right\}, \\ f^{-1}\left(\tfrac{3}{4}\right) &= \{1, -6\}, \\ f^{-1}(v) &= \left\{\tfrac{12}{5}\right\} \quad \textit{with branching}. \end{aligned}$$

*(v) The map factors as*

$$f = \phi \circ q \circ m, \qquad m(z) = \frac{5z}{5z-12}, \quad q(w) = w^2, \quad \phi(s) = \frac{121s - 25}{96s}.$$

*Proof. (i).* Evaluation gives

$$f(0) = \infty, \qquad f(\infty) = 1, \qquad f(1) = \frac{4+5-6}{4} = \frac{3}{4},$$

and

$$f\left(\tfrac{3}{4}\right) = \frac{\frac{9}{4} + \frac{15}{4} - 6}{\frac{9}{4}} = 0,$$

so the claimed 4-cycle holds.

*(ii).* In the local coordinate $w = 1/f$ at $f(0) = \infty$,

$$w(z) = \frac{4z^2}{4z^2 + 5z - 6},$$

which has a zero of order 2 at $z = 0$. Therefore 0 has local degree 2: it is a critical point with critical value $\infty$. Away from 0 we differentiate:

$$f'(z) = \frac{(8z+5)4z^2 - (4z^2+5z-6)8z}{16z^4} = \frac{12 - 5z}{4z^3}.$$

Hence the only finite critical point other than 0 is $z = 12/5$. Its value is

$$f\left(\tfrac{12}{5}\right) = \frac{\left(\frac{48-15}{5}\right)\left(\frac{12+10}{5}\right)}{4 \cdot \frac{144}{25}} = \frac{\frac{33}{5} \cdot \frac{22}{5}}{\frac{576}{25}} = \frac{726}{576} = \frac{121}{96} = v.$$

Finally, $\infty$ is not critical: with $u = 1/z$ near $\infty$ and target coordinate $W = f(1/u) - 1$ near $f(\infty) = 1$,

$$W(u) = \frac{4/u^2 + 5/u - 6}{4/u^2} - 1 = \frac{5}{4}u - \frac{3}{2}u^2,$$

whose derivative at $u = 0$ is $5/4 \neq 0$. Therefore $V(f) = \{\infty, v\}$.

*(iii) and (v).* First verify $\tau^2 = \mathrm{id}$:

$$\tau(\tau(z)) = \frac{6 \cdot \frac{6z}{5z-6}}{\frac{5\cdot 6z}{5z-6} - 6} = \frac{\frac{36z}{5z-6}}{\frac{30z-6(5z-6)}{5z-6}} = \frac{36z}{36} = z.$$

Next, since

$$4\tau^2 + 5\tau - 6 = \frac{144z^2 + 180z - 216}{(5z-6)^2}, \qquad 4\tau^2 = \frac{144z^2}{(5z-6)^2},$$

we have

$$f(\tau(z)) = \frac{144z^2 + 180z - 216}{144z^2} = 1 + \frac{5}{4z} - \frac{3}{2z^2} = f(z),$$

which proves $f \circ \tau = f$. For the factorization, if $w = m(z) = 5z/(5z-12)$, then $q(w) = w^2 = 25z^2/(5z-12)^2$ and

$$\phi(q(m(z))) = \frac{121 \cdot \frac{25z^2}{(5z-12)^2} - 25}{96 \cdot \frac{25z^2}{(5z-12)^2}} = \frac{25\big(121z^2 - (5z-12)^2\big)}{2400z^2}.$$

Since $(5z-12)^2 = 25z^2 - 120z + 144$, the numerator is $25(96z^2 + 120z - 144)$, so

$$\phi(q(m(z))) = \frac{2400z^2 + 3000z - 3600}{2400z^2} = 1 + \frac{5}{4z} - \frac{3}{2z^2} = f(z).$$

*(iv).* We compute the fibers directly.

$$f(z) = 0 \iff (4z-3)(z+2) = 0 \iff z \in \left\{\tfrac{3}{4}, -2\right\},$$

$$f(z) = 1 \iff 4z^2 + 5z - 6 = 4z^2 \iff z = \tfrac{6}{5},$$

and additionally $z = \infty$ satisfies $f(\infty) = 1$, so $f^{-1}(1) = \{\frac{6}{5}, \infty\}$. Next,

$$f(z) = \tfrac{3}{4} \iff 4(4z-3)(z+2) = 12z^2 \iff z^2 + 5z - 6 = 0 \iff z \in \{1, -6\},$$

so $f^{-1}(\frac{3}{4}) = \{1, -6\}$. For the value $v$, we have

$$f(z) = v = \tfrac{121}{96} \iff 384z^2 + 480z - 576 = 484z^2,$$

hence

$$100z^2 - 480z + 576 = 0 \iff (5z-12)^2 = 0,$$

so $f^{-1}(v) = \{12/5\}$ with multiplicity 2. Finally, $f^{-1}(\infty) = \{0\}$ because $4z^2 = 0$ is the only solution of the vanishing denominator, with multiplicity 2. □

*Remark* 4.2. From Lemma 4.1,

$$f^{-1}(B) = \left\{\tfrac{3}{4}, -2,\ 0,\ \tfrac{6}{5},\ \infty,\ 1,\ -6,\ \tfrac{12}{5}\right\},$$

so that

$$A \subset f^{-1}(B), \qquad f^{-1}(B) \setminus A = \left\{-2,\ \tfrac{6}{5},\ -6,\ \tfrac{12}{5}\right\}.$$

In particular, $f$ restricts to an unramified double cover $f : \mathbb{S}^2 \setminus f^{-1}(B) \to \mathbb{S}^2 \setminus B$ with deck group $\langle\tau\rangle$.

## 4.1 The preimage of a $v$-pair disk

**Lemma 4.3.** *Let $a \in \{0, 1, \frac{3}{4}\}$ and let $\gamma = \gamma_{a,v}$ be a simple closed curve bounding a disk $D$ with $D \cap B = \{a, v\}$. Then $\delta = f^{-1}(\gamma)$ is a single essential simple closed curve, the restriction $f : \delta \to \gamma$ has degree 2, and the disk $f^{-1}(D)$ bounded by $\delta$ contains the branch point $12/5$ and exactly one point of $A$.*

*Proof.* By Lemma 4.1, the branch values of $f$ are exactly $\infty$ and $v$. The disk $D$ contains exactly one branch value, namely $v$, and exactly one non-branch value $a$. Therefore the total ramification of $f$ over $D$ is 1: the unique simple branch point $12/5$ maps to $v$.

The monodromy of the degree-two cover $f$ around the boundary $\partial D = \gamma$ is the product of the local monodromies about the marked points lying in $D$. Since $v$ is a branch value of $f$, the local monodromy about $v$ is the nontrivial transposition of the two sheets, whereas the local monodromy about $a$ is trivial. Hence the total monodromy is nontrivial, and $f^{-1}(\gamma)$ is connected: it is a single simple closed curve $\delta$, and $f : \delta \to \gamma$ is a degree-2 covering.

Applying the Riemann–Hurwitz formula to the proper degree-two map $f : f^{-1}(D) \to D$ over the disk $D$, with one branch point of multiplicity 2 over $v$, gives

$$\chi\big(f^{-1}(D)\big) = 2\,\chi(D) - 1 = 2 \cdot 1 - 1 = 1.$$

Since $f^{-1}(D)$ is connected and has a single boundary component $\delta$, it is a disk.

Finally, we determine the $A$-points lying in $f^{-1}(D)$. Since $D \cap B = \{a, v\}$ and $A \subset B$, any point of $A$ lying in $f^{-1}(D)$ must map into $D \cap A = \{a\}$, so

$$f^{-1}(D) \cap A \subset f^{-1}(a).$$

By Lemma 4.1, each fiber over $a \in \{0, 1, \frac{3}{4}\}$ has exactly two points, one in $A$ and one not in $A$:

$$f^{-1}(0) = \{\tfrac{3}{4}, -2\}, \qquad f^{-1}(1) = \{\tfrac{6}{5}, \infty\}, \qquad f^{-1}(\tfrac{3}{4}) = \{1, -6\}.$$

The point $12/5 \in f^{-1}(D)$ is a branch point and lies over $v \notin A$, so it is not an $A$-point. Therefore $f^{-1}(D) \cap A$ consists of the unique $A$-point inside the fiber $f^{-1}(a)$; explicitly it is $\frac{3}{4}$ for $a = 0$, $\infty$ for $a = 1$, and 1 for $a = \frac{3}{4}$. □

## 4.2 The squared twist is special liftable

**Lemma 4.4.** *Let $a \in \{0, 1, \frac{3}{4}\}$ and $\gamma = \gamma_{a,v}$. Then $T_\gamma^2$ is $f$-liftable, with a lift equal to the full twist $T_\delta$ about $\delta = f^{-1}(\gamma)$. Moreover $\Phi_f(T_\gamma^2) = 1 \in \mathrm{Mod}_A$ and $\Phi_A^B(T_\gamma^2) = 1 \in \mathrm{Mod}_A$.*

*Proof.* By Lemma 4.3, $f^{-1}(\gamma) = \delta$ is a single curve mapping to $\gamma$ with degree 2. Choose annular coordinates $(r, \theta)$ near $\gamma$ and $(r, \theta')$ near $\delta$ in which the covering is $(r, \theta') \mapsto (r, 2\theta')$. A representative of $T_\gamma^2$ has the form

$$(r, \theta) \longmapsto (r, \theta + 4\pi\,\rho(r))$$

for a cut-off function $\rho$ supported in the annulus and changing by one full twist across it. Its lift is

$$(r, \theta') \longmapsto (r, \theta' + 2\pi\,\rho(r)),$$

which is precisely the full Dehn twist $T_\delta$ along $\delta$. Hence $T_\gamma^2 \in L_f$, with $\Phi_f$-image realised by $T_\delta$.

We now determine $\Phi_f(T_\gamma^2) = 1$. The lift $T_\delta \in \mathrm{Mod}_{f^{-1}(B)}$ acts by twisting about $\delta$. By Lemma 4.3, after forgetting the points of $f^{-1}(B) \setminus A$, the curve $\delta$ bounds a disk containing exactly one point of $A$. Any homeomorphism of a disk fixing a single interior marked point and equal to the identity on the boundary is isotopic relative to the boundary and the marked point to the identity (the pure mapping class group of a disk with one marked point is trivial; this is the Alexander trick). Hence the class of $T_\delta$ in $\mathrm{Mod}_A$ is trivial, and

$$\Phi_f(T_\gamma^2) = 1 \in \mathrm{Mod}_A\,.$$

For the forgetful image $\Phi_A^B(T_\gamma^2)$, note that after forgetting the point $v$ from the marked set, the curve $\gamma$ bounds a disk whose intersection with $A$ is exactly $\{a\}$. By the same Alexander-trick argument, $T_\gamma$ and hence $T_\gamma^2$ is trivial in $\mathrm{Mod}_A$:

$$\Phi_A^B(T_\gamma^2) = 1 \in \mathrm{Mod}_A\,.$$

□

**Theorem 4.5.** *For each $a \in \{0, 1, \frac{3}{4}\}$, let $\gamma_{a,v}$ be a simple closed curve separating $\{a, v\}$ from the other three points of $B$. Then*

$$T_{\gamma_{a,v}}^2 \in S_f,$$

*and consequently*

$$T_{\gamma_{a,v}}^2 \cdot \mathcal{D}_f = \mathcal{D}_f.$$

*In particular, $\gamma = \gamma_{0,v}$ satisfies the preservation requirement of the original problem with power 2.*

*Proof.* Let $\gamma = \gamma_{a,v}$. By Lemma 4.4, $T_\gamma^2 \in L_f$ and both $\Phi_f(T_\gamma^2)$ and $\Phi_A^B(T_\gamma^2)$ equal the identity in $\mathrm{Mod}_A$. Hence $T_\gamma^2 \in S_f = \{g \in L_f : \Phi_f(g) = \Phi_A^B(g)\}$. Applying Proposition 2.1 with $g = T_\gamma^2 \in S_f$ gives

$$T_\gamma^2 \cdot \mathcal{D}_f = \mathcal{D}_f.$$

Taking $a = 0$ gives the explicit curve $\gamma_{0,v}$. □

# 5 The vanishing-cycle twist lies in $E_f$

We now turn to the action on components. We will show that the squared twist $T_{\gamma_{0,v}}^2$ lies in the subgroup $E_f$ coming from loops in the Hironaka–Koch parameter space $V$, and hence fixes the base component of $\mathcal{D}_f$.

**Lemma 5.1.** *In the Hironaka–Koch parameter space $V = \mathrm{Per}_4(0)^*$, the collision $v = c_1$ occurs as a removed puncture. In the coordinates of [1, §4.1], with the periodic critical point normalized to $c_1 = 0$, the collision condition is equivalent to the conic*

$$G:\ -x^2 + y + 2x - 1 = 0.$$

*In those coordinates the one-parameter family $V$ is represented by the diagonal $x = y$ with the collision and degeneracy loci removed, so*

$$G \cap V : \quad x^2 - 3x + 1 = 0,$$

*and the collision punctures are*

$$x = \frac{3 \pm \sqrt{5}}{2}.$$

*The associated vanishing cycle transported to the base fibre is $\gamma_{0,v}$ under the Hironaka–Koch marking convention.*

*Proof.* This is the coordinate computation of [1, §4.1]. In those coordinates the periodic critical point is normalized to $c_1 = 0$, and the free critical value is a rational function $z(x, y)$. Hironaka–Koch write the collision condition as

$$z(x, y) = 0 \quad \Longleftrightarrow \quad -x^2 + y + 2x - 1 = 0,$$

i.e. the collision conic is $y = (x - 1)^2$. The one-parameter family $V$ is given in this chart by the diagonal $x = y$, after removing the finite set on which the five marked points fail to be distinct or the rational map degenerates. Substituting $y = x$ into $G$ gives

$$-x^2 + x + 2x - 1 = 0 \quad \Longleftrightarrow \quad x^2 - 3x + 1 = 0,$$

with roots $x = (3 \pm \sqrt{5})/2$. At these parameters the two marked points 0 and $v$ coincide, so the marked sphere is not in Teich($B$); such parameters are punctures of $V$. A small loop around either puncture lies in $V$, and the colliding pair is $\{0, v\}$; therefore the vanishing cycle transported to the base fibre is, by definition of $\gamma_{0,v}$, the curve separating $\{0, v\}$ from the remaining three points of $B$. □

**Theorem 5.2.** *With the Hironaka–Koch base point $*_V$ and marking conventions, the squared vanishing-cycle twist satisfies*

$$T_{\gamma_{0,v}}^2 \in E_f.$$

*Proof.* Let $p$ denote one of the collision punctures

$$x = \frac{3 \pm \sqrt{5}}{2}$$

from Lemma 5.1, and let $U$ be a small coordinate disk about $p$ inside the diagonal with local holomorphic coordinate $t$ centred at $p$, so that $U \setminus \{p\} \subset V$. The local meridian $t = \varepsilon e^{2\pi i\theta}$, $0 \le \theta \le 1$, together with the Hironaka–Koch transport path in $V$ from $U$ to the base point $*_V$, defines a based loop $\ell$ in $V$.

The collision has multiplicity two. This is the coordinate-invariant content of Theorem 3.1: in a one-parameter model of the collision, the relative coordinate between the periodic critical point and the free critical value is a nonzero holomorphic function times $t^2$. The normal-form family of Theorem 3.1 and the diagonal in the Hironaka–Koch chart are two holomorphic parametrizations of the same one-parameter locus $\mathrm{Per}_4(0)$ near the collision. The vanishing order of the relative coordinate of two marked points at a puncture is

invariant under holomorphic reparametrization of the parameter and under Möbius changes of coordinate on the sphere. Hence the order-two vanishing proved in Theorem 3.1 also holds for the local parameter $t$ on the Hironaka–Koch diagonal. After possibly shrinking $U$, we may therefore write

$$w(t) = u(t)\,t^2, \qquad u(0) \neq 0,$$

for the relative coordinate of the colliding marked points 0 and $v$. As $t$ goes once around 0, $w(t)$ winds twice around 0.

Inside a small disk containing only the two colliding marked points, the resulting motion is a pure braid with relative winding number two; by the standard braid-to-mapping-class identification in a twice-punctured disk (Farb–Margalit [2, Chapter 9, Theorem 9.1 and §9.1]), this is the square of the Dehn twist about the curve surrounding the two punctures, up to orientation. The local vanishing cycle transported to the base fibre is $\gamma_{0,v}$ by Lemma 5.1.

The exact identification of the based meridian's monodromy uses the Hironaka–Koch base point and marking conventions: the base fibre over $*_V$ is identified with $(\mathbb{S}^2, B)$, the colliding pair is $\{0, v\}$, and the transport path from $*_V$ to the puncture is the one used in that marking. Under this convention the vanishing cycle transported to the base fibre is the specific curve $\gamma_{0,v}$. Thus the based monodromy of $\ell$ is $T_{\gamma_{0,v}}^{\pm 2}$; since $E_f$ is a subgroup, this implies $T_{\gamma_{0,v}}^2 \in E_f$. □

**Corollary 5.3.** *Let $D_0$ be the connected component of $\mathcal{D}_f$ containing the base point $\tau_*$. Then*

$$T_{\gamma_{0,v}}^2(D_0) = D_0.$$

*Proof.* Combine Theorem 5.2 with Lemma 2.2. □

*Remark* 5.4. The arguments above do not show that $E_f$ is normal in $S_f$. Therefore Corollary 5.3 rigorously fixes only the base component. It does not by itself determine the action of $T_{\gamma_{0,v}}^2$ on all components of $\mathcal{D}_f$, and it does not rule out nontrivial behaviour on other components. However, no argument in this paper establishes an infinite orbit for this Dehn twist.

# 6 Consequences and open points

The original problem asks for an explicit simple closed curve $\gamma$ such that a power of $T_\gamma$ preserves $\mathcal{D}_f$ and acts on $\pi_0(\mathcal{D}_f)$ with an infinite orbit. The results above give the following partial answer. The preservation requirement is solved by the explicit curve

$$\gamma = \gamma_{0,v}$$

with power 2: Theorem 4.5 proves $T_{\gamma_{0,v}}^2 \in S_f$, and Proposition 2.1 then gives

$$T_{\gamma_{0,v}}^2 \cdot \mathcal{D}_f = \mathcal{D}_f.$$

More generally, for each $a \in \{0, 1, \frac{3}{4}\}$, the squared twist about $\gamma_{a,v}$ lies in $S_f$ and preserves $\mathcal{D}_f$.

The infinite-orbit requirement is not proved for this Dehn twist. Theorem 5.2 shows

$$T^2_{\gamma_{0,v}} \in E_f,$$

so the base component $D_0$ of $\mathcal{D}_f$ is fixed by Corollary 5.3. The known Hironaka–Koch proof that $\mathcal{D}_f$ has infinitely many components uses the braid element

$$g = s_2\, c\, r_1\, s_1\, c\, r_2,$$

whose image under the Hironaka–Koch quotient map has infinite order; see [1, Theorem 1.3 and Remark 4.7]. The arguments of this paper do not identify this element, or any power of it, with a Dehn twist about a single simple closed curve on $(\mathbb{S}^2, B)$. The infinite orbit produced by $g$ is therefore not realised here by a Dehn twist.

The precise open points are the following.

(1) No argument in this paper shows that a power of $T_\gamma$ for some single essential simple closed curve $\gamma \subset (\mathbb{S}^2, B)$ lies in $S_f \setminus E_f$, which is the standard route in the Hironaka–Koch framework to obtaining nontrivial component orbits.

(2) The canonical curve $\gamma_{0,v}$ does not supply the infinite orbit: its square lies in $E_f$ and fixes the base component $D_0$.

(3) Since $E_f$ is not known here to be normal in $S_f$, membership of $T^2_{\gamma_{0,v}}$ in $E_f$ controls the base component only. Ruling out nontrivial orbits on all components would require additional information, such as normality of $E_f$ or a full coset calculation.

(4) For the other squared $v$-pair twists $T^2_{\gamma_{1,v}}$ and $T^2_{\gamma_{3/4,v}}$, preservation is proved by Theorem 4.5, but their membership or non-membership in $E_f$ is not settled here.

(5) The Hironaka–Koch infinite-component witness $g = s_2 c r_1 s_1 c r_2$ is a braid element; identifying it, or a power of it, with a Dehn twist about an explicit simple closed curve remains open.

(6) A complete classification of essential simple closed curves whose Dehn-twist powers lie in $S_f \setminus E_f$ is not given here.

# References


[1] E. Hironaka and S. Koch, *A disconnected deformation space of rational maps*, arXiv:1602.07378, 2016.

[2] B. Farb and D. Margalit, *A Primer on Mapping Class Groups*, Princeton Mathematical Series, Princeton University Press, 2012.